

Invisible Debt Trap Analytics

An AI-Based Early Warning System for Consumer Credit Rotation
Risk and Debt Carry-Over Pattern Detection in Australia

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The Hidden Problem — Paying Monthly Doesn't Mean Financially Safe

A consumer makes a payment every month — yet their debt never decreases.

Minimum Payments

Paying only the minimum keeps balances alive — interest erodes every repayment

High Utilisation

Balances near the credit limit month after month signal a trapped borrower

No Early Alert

No alarm fires. Not for the consumer. Not until formal delinquency is reached.

Australian Context — The Scale of the Problem

48.9%

of credit card balances
accruing interest (RBA, 2026)

13.5%

of consumers show
problem debt signs (ASIC, 2024)

169K+

National Debt Helpline
contacts in 2024 (+12%)

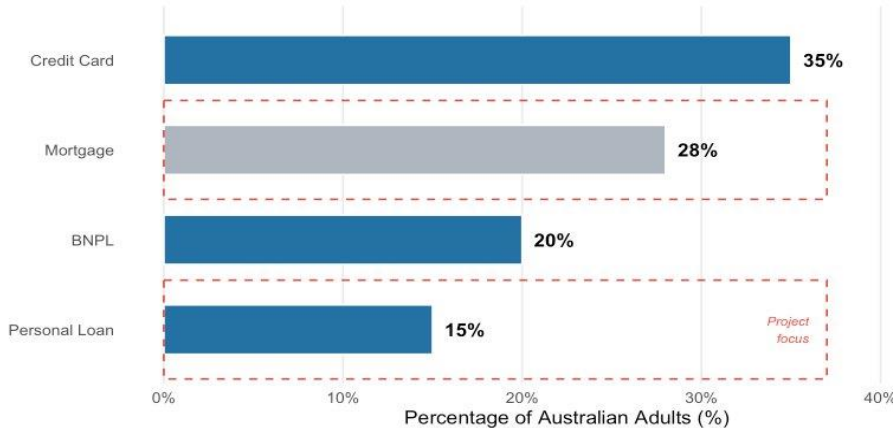
Jun 2025

BNPL now regulated
under NCCP Act (ASIC, 2025)

Debt Type Prevalence Among Australian Adults (NAB Q4 2025)

Debt Type Prevalence Among Australian Adults

Percentage of Australians currently holding each debt type



Source: NAB Australian Wellbeing Survey Q4 2025

Key Observations

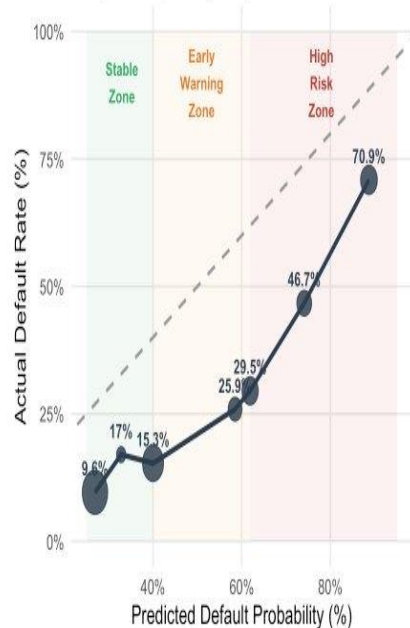
- 35% of Australians carry credit card debt
- 20% hold BNPL obligations
- 15% have personal loans
- Many hold all three simultaneously
- Household debt rose to \$3.31 trillion

The Gap — Existing Systems Are Lender-Facing or Reactive

Invisible Debt Trap Analytics — Detection Gap Evidence

Model Calibration — Predicted vs Actual

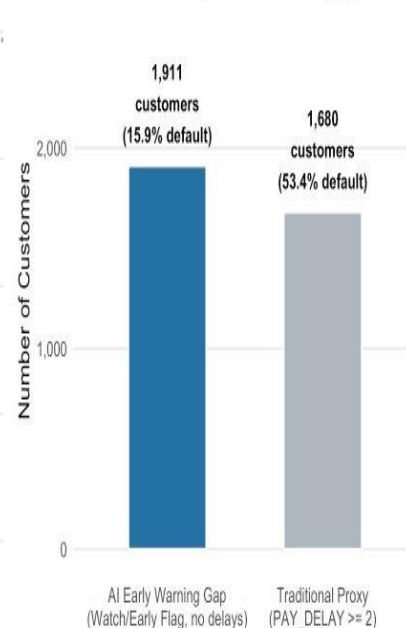
Each point = one probability band | Point size = number of customers



Dashed line = perfect calibration | Proxy = PAY_DELAY_COUNT >= 2 | AI gap = Watch/Early Flag with zero delays

Detection Gap — What Traditional System

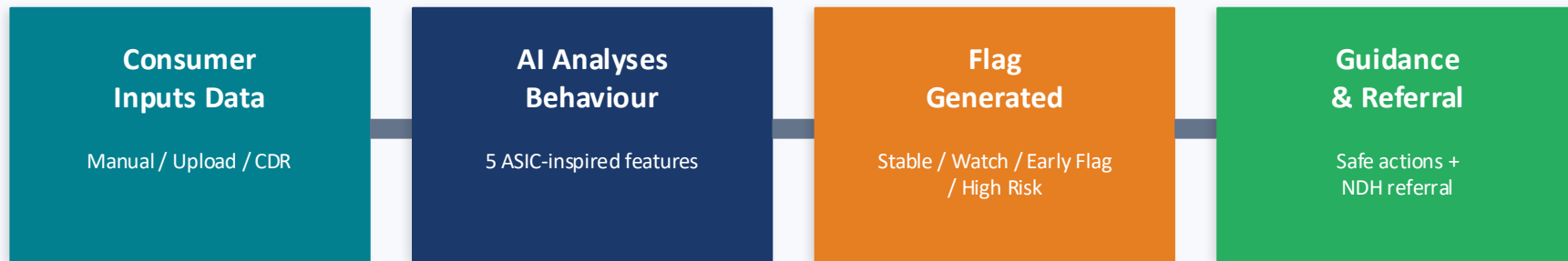
Blue = customers our AI system catches before any payment delay o



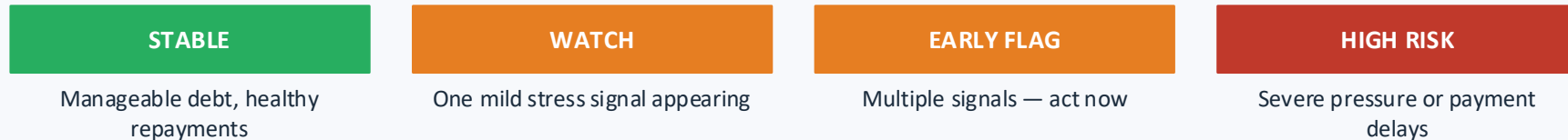
Tool / Approach	What It Does	What It Misses
ASIC REP 788	Identifies problem debt at lender level	Consumer never sees the flag
UK FCA PS18/4	Mandates lender intervention	Triggers only after 18 months
Frollo / WeMoney	Aggregates accounts via CDR	No pre-delinquency AI flag
Accumn.ai	AI early warning — for lenders	Lender sees it, consumer does not
Ala'raj et al. 2021	Bi-LSTM credit behaviour model	Institutional only — no consumer output

Detection Gap Evidence — Our R Analysis (UCI Dataset, n = 8,989)

Proposed Solution — Invisible Debt Trap Analytics



The Four Risk Categories



Core principle: Consumer-controlled data. Bank has no access to the flag. All guidance is general information only — no personal financial advice. Free for high-risk users.

The Data — Two Separate Evidence Layers

UCI Credit Card Dataset

Primary ML Modelling Dataset

- Yeh & Lien (2009) | DOI: 10.24432/C55S3H
- 30,000 individual credit card records
- 6 months of repayment behaviour
- 24 features + binary default target
- CC BY 4.0 licence — freely available
- Used in peer-reviewed papers in 2024 & 2025

RBA Statistical Table C1

Australian Context Layer

- Reserve Bank of Australia — official government
- Monthly data: January 1985 to March 2026
- 135 months analysed (2015–2026)
- Revolving ratio = interest balances / total balances
- Latest: 48.9% revolving (March 2026)
- Freely downloadable at rba.gov.au
-  Australian, authorised, live, updated monthly

Method — Feature Engineering, Data Wrangling and ML Models

Data Wrangling & Feature Engineering

- Removed 36 duplicate rows
- Recoded undocumented EDUCATION & MARRIAGE values
- Standardised PAY column coding (-2 → -1)
- Outlier audit using IQR method
- Converted dates, standardised column names

5 ASIC-Inspired Proxy Features

- Utilisation Rate — avg bill / credit limit
- Repayment Ratio — avg payment / avg bill
- Minimum Pay Frequency — months $\leq$ 3% of limit
- Balance Persistence — months bill > 80% of limit
- Payment Delay Count — months with any delay

Machine Learning Models

Baseline

Logistic Regression

Interpretable coefficients
| Week 7 FIT5145

Recommended

Classification Tree

Visual decision rules
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Ensemble

Random Forest

200 trees, reduces overfitting
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Production
Extension

SHAP Values

Explains WHY per customer
Beyond current scope

5-fold cross-validation | Class weights 3.52:1 | Recall = priority metric

Results — Classification Tree Recommended on Recall

Model Performance Comparison — All Three Models (Test Set n = 8,989)

Model Performance Comparison — All Three Models

Classification Tree shows highest Recall — most important for debt stress detection



63%

Recall — Class. Tree
(Best for early warning)

6.5x

High Risk vs Stable
default rate ratio

0.766

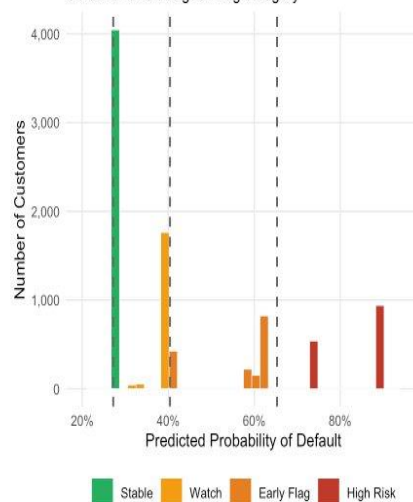
AUC — Logistic Regression
(strong discriminative power)

Key Findings

Invisible Debt Trap Flag — Probability Mapping Validation

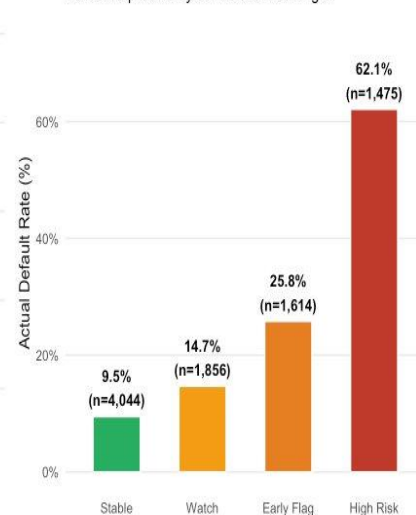
Predicted Default Probability Distribution

Colour shows assigned flag category



Default Rate by Flag Category

Validates probability bands are meaningful



Flag Validation — Default Rate per Category

Business Value and Ethics

Who Benefits

Australian Consumers

Early flag before any missed payment

18–29 Age Group

Highest-risk group — 2× more likely to show problem debt (ASIC)

National Debt Helpline

Receives earlier, better-informed referrals

Vulnerable Populations

Anonymous check — no credit score impact

Ethics & Revenue Model

- Free tier: flag, safety check, NDH referral — always free
- Paid tier: advanced analytics, scenario planning
- B2B future: licensed fintechs and wellness platforms
- No personal financial advice — general guidance only (Corporations Act 2001)
- Privacy Act 1988 + 13 Australian Privacy Principles
- CDR Privacy Safeguards for open banking version
- Fairness audit required — 18–29 age group over-represented in training data

Core rule: Emergency warnings are never behind a paywall.

What This Project Proves

- ✓ 1,911 customers flagged early with zero payment delays — invisible to existing monitoring systems
 - ✓ High Risk customers default at 6.5× the rate of Stable — the flag is genuinely meaningful
 - ✓ 48.9% of Australian credit card balances are currently revolving — the problem is real and worsening
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Future Direction

- CDR open banking integration — real-time Australian consumer data without password sharing
- SHAP explainability — consumer sees exactly why they were flagged
- Verified Debt Relief Support Network — regulated referral pathway for high-risk users